

Character-Resolved Fixed Fibres for a Heisenberg Nilmanifold Automorphism

Route-A structural certificate C151

Abstract

For a frozen Heisenberg lattice automorphism, we derive the central rotation over every horizontal fixed class. Its zero level counts clean fixed circles, and a finite root-of-unity average gives the component count at every iterate. Exact rational histograms through iterate twelve distinguish horizontal fixed classes from actual components. This is a source-side clean-family result, not an isolated-orbit or target construction.

1 Fixed-fibre equation

On $N = \mathbb{Z}^3 \setminus H$, use

$$(x, y, z)(X, Y, Z) = (x + X, y + Y, z + Z + xY), \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

and $\Phi(v, z) = (Av, z + q(v))$, where $q(x, y) = x(x - 1) + xy + y(y - 1)/2$. Put $B = A^n$, $M = B - I$, and $q_n(v) = \sum_{j=0}^{n-1} q(A^j v)$. The horizontal fixed classes are $\mathbb{Z}^2/M\mathbb{Z}^2$. If $m = Mv$, the left-quotient equation is

$$(Bv, z + q_n(v)) = (m_1, m_2, k)(v, z) = (v + m, z + k + m_1 v_2).$$

It follows that the fibre rotation is

$$\rho_n(v) = q_n(v) - m_1 v_2 \pmod{1}, \quad (1)$$

and the fibre is fixed exactly when $\rho_n(v) = 0$. The condition is independent of z , so a zero produces the whole central circle.

For completeness, this is representative invariant. Replace v by $v + r$ with $r \in \mathbb{Z}^2$, put $s = Mr$, and use the automorphism identity

$$q_n(v + r) - q_n(v) - q_n(r) = (Bv)_1(Br)_2 - v_1 r_2.$$

The change in (1) reduces to $q_n(r) + \det(v, s) + m_1 s_2 - s_1 r_2$. Since B preserves area, $\det(v + m, r + s) = \det(v, r)$, and the change becomes

$$q_n(r) - m_1 r_2 + m_2 r_1 + s_1 m_2 - s_1 r_2 \in \mathbb{Z}.$$

Thus (1) is well-defined on the horizontal class. Along a fixed fibre, the derivative is block lower triangular with diagonal blocks B and 1. Since $I - B$ is invertible, $\ker(I - D\Phi^n)$ is exactly the central tangent, so the component is clean.

2 Finite root-of-unity projector

Let $D_n = |\det M|$. The adjugate formula puts the coordinates of v in $D_n^{-1}\mathbb{Z}$, while q_n is quadratic with half-integral coefficients. Thus $\rho_n \in Q_n^{-1}\mathbb{Z}/\mathbb{Z}$ for $Q_n = 2D_n^2$. Root-of-unity orthogonality gives

$$C_n = \frac{1}{Q_n} \sum_{a=0}^{Q_n-1} \sum_{[m] \in \mathbb{Z}^2/M\mathbb{Z}^2} e^{2\pi i a \rho_n(m)}. \quad (2)$$

3 Exact certificate and boundary

The exact pairs (D_n, C_n) through $n = 12$ are

n	1	2	3	4	5	6	7	8	9	10	11	12
D_n	1	5	16	45	121	320	841	2205	5776	15125	39601	103680
C_n	1	1	4	1	21	4	57	1	148	105	397	144

The checker reconstructs every rotation through $n = 12$ by exact direct cocycle iteration; a separate SymPy path rebuilds the first eight full histograms. Early finite patterns are not extrapolated.

The tuple is (A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT). Haar Koopman evolution is only a formal unitary hint. We claim no isolated primitive-orbit determinant, target divisor, target functional equation or counting law, arithmetic/local factor, Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.